AUTOOPS SCHOOL · COURSE STUDY SHEET

DevOps Engineering — Beginner to Practical

Paid course — ¹4,999 · 8 modules · 3 calendar months of access after Admin approval

About this course

Learn practical DevOps concepts, automation tools, CI/CD workflows, containerization, cloud infrastructure, and monitoring through structured learning and hands-on projects.

What you will learn

- Understand the DevOps lifecycle
- Learn Git and source control
- Understand Jenkins and CI/CD pipelines
- Learn Docker and containerization
- Understand Kubernetes fundamentals
- Explore AWS and Azure cloud concepts
- Learn Terraform and Ansible
- Understand Prometheus and Grafana monitoring

Modules in this course

1. Linux Foundations

   The operating-system skills every DevOps engineer relies on daily.

   Topics: Linux refresher: filesystem, permissions, users, Networking basics for engineers: ports, DNS, SSH, Process management and services, Shell scripting for automation

2. Git and Source Control

   Collaborate safely and build repeatable workflows for application changes.

   Topics: Repositories, commits and branching, Remotes, pull requests and code review, Resolving merge conflicts and rebasing, Git workflow strategies for teams

3. Jenkins and Continuous Integration

   Automate build, test and delivery pipelines end to end.

   Topics: Jenkins architecture and setup, Freestyle vs pipeline jobs, Declarative Jenkinsfile walkthrough, Build, test and deploy stages, Credentials, webhooks and notifications

4. Docker and Containerization

   Package applications into portable, consistent containers.

   Topics: Images, containers and the Dockerfile, Layers, caching and multi-stage builds, Docker networking and volumes, Docker Compose for multi-service apps, Registry and image security basics

5. Kubernetes and Container Orchestration

   Deploy, scale and operate containers on a resilient platform.

   Topics: Cluster architecture: control plane and nodes, Pods, deployments, replica sets, Services, ingress and DNS, ConfigMaps, secrets and resource limits, Scaling, health probes and rolling updates

6. Cloud Computing — AWS and Azure

   Core cloud services and deployment patterns across two platforms.

   Topics: Cloud accounts, IAM and regions, Compute: EC2 and Azure VMs, Storage, networking and load balancing, Managed databases and serverless, Deploying a containerised app to the cloud

7. Infrastructure Automation — Terraform and Ansible

   Provision and configure environments with reviewable, repeatable code.

   Topics: Terraform fundamentals: providers, resources, state, Modules, variables and outputs, Planning and applying safely, Ansible inventory, playbooks and roles, Templates, handlers and idempotency, Terraform + Ansible together

8. Monitoring — Prometheus and Grafana

   Measure systems, investigate incidents and improve reliability.

   Topics: Metrics, logs and traces explained, Prometheus architecture and PromQL, Exporters, scraping and alert rules, Alertmanager routing and silences, Grafana dashboards and variables, SLOs, incident response and troubleshooting

How to use this sheet

Sign in at the AutoOps School student portal, open “DevOps Engineering — Beginner to Practical”, and enroll. After Admin approval your 3-month access window starts and every PDF in the course materials section becomes available to view and download.